# Waseem Raza, PhD

## Machine Learning Engineer | Model Training, Inference & Python Systems

Deep Learning | Numerical Computing | Reproducible ML | Evaluation & Performance

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LinkedIn | GitHub | Google Scholar | Portfolio

## Professional Summary

Machine Learning Engineer and researcher with a PhD in Electrical and Computer Engineering and 7+ years of combined industry and research experience developing, training, validating, and debugging machine-learning systems. Strong hands-on background in Python, PyTorch, TensorFlow/Keras, scikit-learn, XGBoost, NumPy, SciPy, Pandas, and Hugging Face Transformers, with experience spanning neural networks, autoencoders, GANs, reinforcement learning, transfer learning, domain adaptation, classification, anomaly detection, time-series analysis, and multimodal data. Built reproducible experimentation and data pipelines across AT&T, Futurewei, university research, and cloud analytics environments, including 100+ controlled ML experiments, 500K+ operational-log analysis, and production-oriented model validation under distribution shift.

## Technical Skills

**Python & Numerical Computing:** Python, NumPy, SciPy, Pandas, SQL, C/C++, Jupyter, vectorized data processing, statistical analysis, numerical experimentation, Linux, Bash/CLI, Git.
**ML & Deep Learning:** PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, Hugging Face Transformers; DNNs, CNNs, LSTMs, autoencoders, GANs, reinforcement learning, transfer learning, domain adaptation, classification, regression, anomaly detection, forecasting.
**Training, Inference & Evaluation:** data preparation, feature engineering, train/validation/test workflows, cross-validation, model comparison, hyperparameter experimentation, batch inference, metric tracking, error analysis, failure-mode assessment, robustness testing, distribution-shift evaluation.
**ML Engineering & Reproducibility:** reusable Python pipelines, experiment automation, deterministic/reproducible workflows, data-quality checks, model-validation harnesses, Git-based development, CI/CD, GitHub Actions, Docker, Kubernetes, OpenShift.
**Data & Compute Platforms:** Azure Databricks, Apache Spark/PySpark, Synapse Analytics, Azure Functions, cloud data pipelines, ETL/ELT, structured/unstructured data processing, Google Cloud ML workflows.
**AI-Assisted Development:** coding-agent-assisted Python development, debugging, test generation, refactoring, repository iteration, and structured review of implementation correctness.

## Professional Experience

**Machine Learning Engineer** **02/2025 – Present**
*AT&T, contractor via Innovatix Inc.* *Bedminster, NJ*

- Own end-to-end development, testing, and operational integration of machine-learning diagnostics for critical E911 service-assurance workflows, improving automated root-cause classification accuracy from 20% to 80%.
- Design domain-adaptive neural models and reusable preprocessing pipelines that improve generalization across heterogeneous operational domains and distribution shifts.
- Build cross-domain model-validation, error-analysis, and failure-mode assessment workflows to identify data/model defects and verify readiness for operational deployment.
- Translate network and public-safety requirements into measurable ML objectives, validated datasets, repeatable experiments, and production-oriented decision-support workflows.

**Data Scientist and AI/ML Engineer** **09/2024 – 12/2024**
*Impac Exploration Services* *Weatherford, OK*

- Built machine-learning and LLM-assisted analytical workflows for large geochemical datasets using Python and Azure cloud services.
- Developed scalable ingestion, cleansing, feature-preparation, transformation, and validation pipelines with Azure Databricks, Azure Functions, and Synapse Analytics.
- Automated ETL workflows, reducing data-preparation time by 40%; implemented cross-validation, data-quality, and consistency checks that improved reliability of analytical outputs by 20%.

**Graduate Research Assistant** **09/2019 – 07/2025**
*AI4Networks, University of Oklahoma* *Tulsa, OK*

- Designed reusable Python simulation and data-generation platforms for large-scale ML experimentation, including a 3GPP-compliant network simulator and AI-enabled 5G testbed.
- Built and evaluated transfer-learning, adversarial-learning, GAN, autoencoder, and reinforcement-learning pipelines for sparse-data and cross-domain modeling.
- Developed robustness-evaluation workflows for distribution shifts, missing data, and limited labeled samples; led 100+ experiments from problem formulation through quantitative validation.

- Developed scalable self-healing and outage-management frameworks that improved modeled network resilience by 30%.

**Advanced Wireless Technology Development Intern** **09/2022 – 01/2023**
*Futurewei Technologies* *Schaumburg, IL*

- Improved autoencoder-based channel-state-information compression by 25% using transfer learning and systematic experimentation on large wireless datasets.
- Conducted 50+ experiments to validate model accuracy, robustness, and generalization under changing data conditions; translated results into recommendations for next-generation communication systems.

**Advanced-Degrees Intern** **06/2024 – 08/2024**
*AT&T Labs* *Austin, TX*

- Analyzed 500K+ LTE/5G-NR diagnostic logs and applied ML methods to detect anomalies, reducing diagnostic time by 20%.
- Improved data-driven decision algorithms through structured experimentation, model evaluation, and validation over large operational datasets.

## Education

**PhD, Electrical and Computer Engineering** *09/2019 – 07/2025*
University of Oklahoma, Norman, Oklahoma, USA

**MSc, Telecommunication Engineering** *08/2014 – 08/2016*
University of Engineering and Technology, Taxila, Pakistan

**BSc, Telecommunication Engineering** *11/2010 – 06/2014*
University of Engineering and Technology, Taxila, Pakistan

## Selected ML Engineering Projects

**Wearable Intelligence for Health Monitoring** *Research Project | 2024–2026*

Developed deep-learning workflows for continuous recognition of medication-taking and dietary activities from smartwatch accelerometer and gyroscope streams. Built preprocessing/windowing pipelines, trained CNN/LSTM-style models, compared sequence lengths and sampling configurations, and evaluated cross-user generalization with recall-oriented metrics.

*Tools & methods:* Python, deep learning, CNN/LSTM, time-series segmentation, feature engineering, smartwatch IMU, train/test evaluation, cross-user validation, confusion matrices, model comparison.

**Propagation Models & Digital Twins** *Research Project | 2024*

Developed robust data-driven propagation models for wireless digital twins using domain-informed neural pretraining, GAN-based data augmentation, and scarce measured data. Evaluated synthetic-data fidelity and downstream regression performance, including robustness under domain shift and limited training data.

*Tools & methods:* Python, deep neural networks, GANs, transfer/domain adaptation, synthetic data, regression, RMSE/$R^2$, statistical validation, 3GPP-informed modeling.

**SyntheticNET: 3GPP-Compliant Network Simulation** *Research Project | 2020–2024*

Developed a Python-based simulation and synthetic-data environment for repeatable large-scale experiments when real measurements are sparse or expensive. Generated controlled mobility, propagation, and network-state datasets used to train, test, and compare ML approaches across consistent scenarios.

*Tools & methods:* Python, numerical simulation, synthetic data generation, 3GPP models, experiment automation, data pipelines, reproducibility, mobility/propagation modeling.

**Autoencoder CSI Compression & Transfer Learning** *Industry Research Project | 2022–2023*

Developed and evaluated autoencoder-based channel-state-information compression workflows for large wireless datasets, using transfer learning to improve reconstruction/compression performance by 25%. Ran 50+ controlled experiments across changing data conditions to quantify accuracy, robustness, and generalization before translating results into engineering recommendations.

*Tools & methods:* Python, autoencoders, deep neural networks, transfer learning, representation learning, model compression for signal representations, train/validation experiments, robustness testing, performance benchmarking.

**Financial ML and Forecasting** *Portfolio Project | 2024–2025*

Developed structured-data ML workflows for credit-risk and loan-default classification alongside time-series forecasting experiments. Implemented data preparation, feature engineering, model comparison, cross-validation, and predictive-performance analysis over finance-oriented datasets.

*Tools & methods:* Python, scikit-learn, classification, time-series forecasting, feature engineering, cross-validation, model evaluation, predictive analytics.

**Cloud Computer Vision** *Guided Project | 2025*

Implemented image-classification, object-detection, and vehicle-counting workflows with managed cloud models and Python inference pipelines. Validated preprocessing, prediction outputs, class/count aggregation, and model behavior across visual inputs.

*Tools & methods:* Python, TensorFlow concepts, computer vision, object detection, image classification, Google Cloud AutoML, preprocessing, inference, model evaluation.

**Agentic AI and RAG Applications** *Portfolio Project | 2026*

Built Python workflows that combine retrieval, vector search, tool-using agents, state/memory, evaluation, and multi-step orchestration. Used iterative agent-assisted development and validation to structure implementation, debug workflows, and verify expected behavior across multi-stage tasks.

*Tools & methods:* Python, LangGraph, LangChain, RAG, vector databases, embeddings, tool calling, agent state/memory, workflow evaluation, coding-agent-assisted iteration.

**Domain-Adaptive E911 Diagnostic ML System** *Industry ML System | 2025–Present*

Developed and operationalized a machine-learning diagnostic system for critical E911 service-assurance workflows, improving automated root-cause classification accuracy from 20% to 80%. Built reusable preprocessing, cross-domain evaluation, error-analysis, and failure-mode assessment workflows to verify correctness and deployment readiness across heterogeneous operational conditions.

*Tools & methods:* Python, supervised classification, text/NLP features, domain adaptation, reusable preprocessing, cross-domain validation, error analysis, failure-mode assessment, production integration, objective performance verification.

## Selected Technical Training

**Machine Learning** *Stanford Online / Coursera | Completed 03/2020*

Regression, classification, regularization, clustering, model evaluation, bias/variance, and practical model-selection workflows for supervised and unsupervised learning.

**Generative AI Engineering and Fine-Tuning Transformers** *IBM / Coursera | Completed 08/2026*

Transformer architectures, pretrained models, task adaptation, fine-tuning strategies, tokenization concepts, training/inference tradeoffs, and LLM evaluation.

**Introduction to Deep Learning & Neural Networks with Keras** *IBM / Coursera | Completed 08/2026*

Neural-network architecture, activation functions, optimization, model training, validation, and iterative model improvement using Keras-based workflows.

**Python for Data Science, AI & Development** *IBM / Coursera | Completed 07/2026*

Python programming, data structures, data manipulation, NumPy/Pandas-oriented analytical workflows, and reusable application development for data science and AI.

**Introduction to Computer Vision with TensorFlow** *Google Cloud / Coursera | Completed 08/2024*

Image preprocessing, convolutional-network concepts, image classification, model construction, training, validation, and practical evaluation with TensorFlow workflows.

## Selected Research & Validation Evidence

**An AI-Driven Framework for Enhancing Resilience in Propagation Models to Enable Digital Twin** *IEEE PIMRC, 2024*

Robust model development and evaluation under scarce measured data and distribution shift using neural pretraining and generative augmentation.

**Real-World Medication Adherence Monitoring Using Hybrid Neural Networks and Smartwatches** *HealthINF, Accepted 2026*

Deep-learning sequence classification, multimodal wearable signals, cross-user evaluation, and real-world reliability analysis.

**Data-Driven Intelligent Outage Management for High Shadowing Environments in 5G&B Networks** *IEEE SmartNets, 2024*

Machine-learning-based anomaly/outage detection, data-driven decision support, and validation under challenging operational conditions.

**On Multi-Parameter Optimization and Proactive Reliability in 5G and Beyond Cellular Networks** *Sensors, 2025*

Predictive modeling, optimization, reliability analysis, and systematic evaluation for data-driven network operation.

## Coding, Review & Reproducibility Practices

- Build Python workflows as testable, repeatable stages for data preparation, model execution, evaluation, error analysis, and result validation rather than ad-hoc notebook-only experiments.
- Use Git, command-line tooling, automated tests, GitHub Actions, and coding-agent-assisted iteration to review implementations, debug failures, refine code, and verify objective correctness.
- Comfortable reviewing ML code and experimental outputs for data leakage, shape/type errors, metric misuse, reproducibility gaps, cross-domain failure modes, and inconsistencies between intended and actual model behavior.
- Evaluate candidate implementations against explicit correctness and performance criteria, documenting assumptions, metrics, failure cases, and reproducibility before accepting results.